

SQL AGGREGATE FUNCTIONS & OPERATORS

EXERCISE 2

Question 1

```
SELECT DISTINCT department  
FROM Students;
```

department
IT
HR
Finance

Question 2

```
SELECT department,  
       AVG (age) AS avg-age  
FROM Students  
GROUP BY department;
```

department	avg-age
IT	21
HR	22
finance	23

Question 3

```
SELECT department,  
       COUNT (*) AS student-count  
FROM Students  
GROUP BY COUNT (*) > 1;
```

department	Student_count
IT	2
HR	2
finance	1

Question 4

```
SELECT student_id, name, age, department
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student-id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	finance

Question 5

```
SELECT student_id, name, age, department
FROM Students
WHERE department = 'IT' OR
      department = 'HR' AND
      age > 21;
```

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

```
SELECT department,  
       sum(credit) AS total-credits  
FROM courses COurses  
GROUP BY department  
HAVING sum(credit) > 5;
```

department	total-credits
IT	11

Question 7

```
SELECT course-id, course-name, department, credits  
FROM courses  
WHERE NOT credits = 4;
```

course-id	course-name	department	credits
101	SQL-Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

```
SELECT course-id, course-name, credits  
FROM courses  
ORDER BY credits Desc  
Limit 3;
```

course-id	course-name	credits
102	IT	4
103	IT	4
101	IT	3

Question 9

SELECT max (grade) AS max-grade,
min (grade) AS min-grade,
Avg. (grade) AS Avg-grade
FROM enrollments;

max-grade	min-grade	Avg-grade
90	78	85

Question 10

SELECT course-id,
COUNT (*) AS enrollment-count
FROM enrollments
GROUP BY course-id;

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Question 11

SELECT department,
Sum (salary) AS total-salary,
Sum (Bonus) AS total-Bonus
FROM Salaries
GROUP BY department;

department	total - Salary	total - Bonus
IT	122 000	10500
HR	104 000	7500
finance	70 000	6000

Question 12

```

SELECT department,
       Avg (Salary) AS Avg - Salary
FROM   Salaries
GROUP BY department
HAVING Avg (Salary) > 50 000 ;

```

department	Avg - Salary
Finance	70 000
IT	61 000

Question 13

```

SELECT employee-id, name, salary, bonus,
       (Salary + bonus) AS total - Compensation
FROM   Salaries
WHERE  (Salary + bonus) > 60 000 ;

```

employee-id	name	Salary	bonus	total - Compensation
1	Tom	60000	5000	65 000
3	Spike	70000	6000	76 000
4	Tyke	62000	5500	67 500

Question 14

SELECT department,
SUM (budget) AS total-budget,
AVG (budget) AS avg-budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70 000;

department	total-budget	avg-budget
IT	210 000	135 000
finance	80 000	80 000

Question 15

SELECT project-id, project-name, department, budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department != 'marketing';

Project-id	project-name	department	budget
1	AI APP	IT	120 000
2	payroll system	finance	80 000
5	HR portal	HR	50 000